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1 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

At a glance

[P1] Constructing a Regional Slang Lexicon for Electricity Demands via Soft-Label Bidirectional Anchoring and Active Learning · relevance 0.22
A weak-supervision pipeline builds a high-precision Yuxi-dialect slang lexicon for power-grid customer service, achieving perfect dialect precision but yielding a modest-sized lexicon and limited generalizability.

[P1] Constructing a Regional Slang Lexicon for Electricity Demands via Soft-Label Bidirectional Anchoring and Active Learning

Rong Chang, Lijing Shi, Zhaoxiang Zang · arXiv 2611.04231v1 · https://arxiv.org/abs/2611.04231v1 · relevance 0.22

TL;DR: A weak-supervision pipeline builds a high-precision Yuxi-dialect slang lexicon for power-grid customer service, achieving perfect dialect precision but yielding a modest-sized lexicon and limited generalizability.

Problem
Power-grid customer service increasingly relies on community social media where users describe outages and faults in informal regional dialect, producing non-standard morphology and out-of-vocabulary expressions that make manual lexicon construction costly and degrade conventional NLP tools[1].

Method
The authors propose a weak-supervision framework that (1) extracts bigram candidates via Pointwise Mutual Information anchored on 16 Yuxi dialect markers with bidirectional soft-label power-domain layering, (2) filters candidates with a four-condition fragment filter, (3) scores retained candidates using a dual-channel classifier that combines Sentence-BERT semantic similarity to 65 power-domain seed terms with a jieba-based wordhood measure, and (4) applies uncertainty-aware active learning with three query strategies (Margin, TD-CAL, Random) and two oracle types (rule-based morphological matching and a retrieval-augmented LLM)[2][4]. Evaluation uses a higher-capability LLM calibrated against human expert annotations via Cohen's ?[2][3].

Key results
- The evaluator-estimated dialect precision (DP) is 1.0 with zero false positives[1][5].
- Soft-label layering expands the candidate pool by 751 % (from 41 to 349 terms) while preserving the same DP[1][7].
- Human calibration on 46 annotations yields ?_dialect = 1.0 and ?_power = 0.85[1][2].
- The final lexicon contains 25-33 terms (4-5 of which are power-relevant) across configurations[6][7].
- Candidate generation extracted 349 candidates (347 bigrams, 2 unigrams); the fragment filter removed 291 (83.4 %) leaving 58 for classification[5].
- The SBERT+wordhood classifier accepted 19 candidates (score >= 0.78) and placed 39 in an uncertain pool (0.45 <= score < 0.78)[5].
- All six configurations achieve DP = 1.0[5].
- Under the high-acceptance LLM oracle (~89 % acceptance) all configurations converge to 4 power-dialect terms, whereas the rule-based oracle shows modest differentiation[2].
- Ablation shows soft-label layering expands the final lexicon by 213 % relative to strict power-gating[7].

Limitations
The study is confined to a single dialect (Yuxi), domain (power grid), and platform (WeChat), with no empirical transfer tests[6]. It reports only precision-oriented metrics; recall against a gold lexicon and full human evaluation beyond the 46-item calibration set are absent[6]. The 11,177-message corpus may not

capture the full diversity of dialect expressions[6]. No comparisons with established phrase-mining baselines are provided, and the active-learning experiment is underpowered (only 39 uncertain candidates and a single random seed)[6]. The evaluator shares the same model family as the selection oracle, and cross-model validation has not been executed[6]. Downstream task impact (e.g., complaint classification) is not assessed, and many pipeline hyperparameters were tuned without a held-out validation set, leaving their sensitivity unanalyzed[6].

Sources

[1] arXiv 2611.04231v1.pdf, arXiv abstract

[2] arXiv 2611.04231v1.pdf, page 5

[3] arXiv 2611.04231v1.pdf, page 9

[4] arXiv 2611.04231v1.pdf, page 14

[5] arXiv 2611.04231v1.pdf, page 18

[6] arXiv 2611.04231v1.pdf, page 25

[7] arXiv 2611.04231v1.pdf, page 26